

CONTINUITY



SYNOPSIS

Continuity at a point :

- A function f is said to be continuous at ' a ' if f is defined in a neighbourhood of ' a ' and

$$\lim_{x \rightarrow a} f(x) = f(a).$$

- i) If $\lim_{x \rightarrow a^-} f(x) = f(a)$ then $f(x)$ is left continuous at $x = a$.
- ii) If $\lim_{x \rightarrow a^+} f(x) = f(a)$ then $f(x)$ is right continuous at $x = a$.

W.E-1: The function $f(x) = \frac{(3^x - 1)^2}{\sin x \cdot \ln(1+x)}$, $x \neq 0$ is continuous at $x = 0$. Then the value of $f(0)$ is

Sol: Given $f(0) = \lim_{x \rightarrow 0} \frac{(3^x - 1)^2}{\sin x \cdot \ln(1+x)}$

$$\Rightarrow f(0) = \lim_{x \rightarrow 0} \frac{\left(\frac{3^x - 1}{x}\right)^2}{\left(\frac{\sin x}{x}\right) \cdot \left(\frac{\ln(1+x)}{x}\right)} = (\ln 3)^2$$

- A function f is said to be continuous in an open interval (a, b) if it is continuous at each and every point in the interval (a, b) .
- A function f is said to be continuous on $[a, b]$ if
- f is continuous at each point of (a, b)
 - f is right continuous at $x = a$
 - f is left continuous at $x = b$

W.E-2: Let f be a continuous function on $[1, 3]$. If f takes only rational values for all x and $f(2) = 10$ then $f(1.5)$ is equal to

Sol: $f(x)$ is continuous function on $[1, 3]$ and takes only rational values then $f(x)$ is constant function.

$$\therefore f(2) = f(1.5) = 10$$

Discontinuity :

- If $f(x)$ is not continuous at $x = a$, we say that $f(x)$ is discontinuous at $x = a$.

$f(x)$ will be discontinuous at $x = a$ in any of the following cases :

- $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exist but are not equal.
- $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ exist and are equal but not equal to $f(a)$
- $f(a)$ is not defined
- At least one of the limit doesn't exist.

W.E-3: Let $f(x)$ be defined in the interval $[0, 4]$

such that $f(x) = \begin{cases} 1-x, & 0 \leq x \leq 1 \\ x+2, & 1 < x < 2 \\ 4-x, & 2 \leq x \leq 4 \end{cases}$ then

number of points where $f(f(x))$ is discontinuous is

Sol: $f(x)$ is discontinuous at $x = 1$ and $x = 2$

$\Rightarrow f(f(x))$ is discontinuous when $f(x) = 1$ & 2

Now $1-x=1 \Rightarrow x=0$, where $f(x)$ is continuous

$$x+2=1 \Rightarrow x=-1 \notin (1, 2)$$

$$4-x=1 \Rightarrow x=3 \in [2, 4]$$

Now, $1-x=2 \Rightarrow x=-1 \notin [0, 1]$

$$x+2=2 \Rightarrow x=0 \notin (1, 2)$$

$$4-x=2 \Rightarrow x=2 \in [2, 4]$$

Hence, $f(f(x))$ is discontinuous at two points, $x = 2, 3$.

- If f and g are continuous functions of x at $x = a$, then the following functions are continuous at $x = a$.

CONTINUITY

i) $f + g$ ii) $f - g$ iii) $f \cdot g$ iv) cf

if $c \in \mathbb{R}$ v) $\frac{f}{g}$ if $g(a) \neq 0$.

Note: Even if f and g are not continuous at $x = a$,

$f \pm g, f \cdot g, \frac{f}{g}, g \circ f$ may be continuous at $x = a$.

- If f is continuous at $x = a$ and g is continuous at $f(a)$, then $(g \circ f)$ is continuous at $x = a$.
- If f is continuous in $[a, b]$ then it is bounded in $[a, b]$. i.e. there exist k and m such that $k \leq f(x) \leq m, \forall x \in [a, b]$ where k and m are minimum and maximum values of $f(x)$ respectively in the interval $[a, b]$.
In this case f takes every real value between k and m at least once. Thus range of f is $[k, m]$.
- If f is continuous on $[a, b]$ such that $f(a)$ and $f(b)$ are of opposite signs then there exist at least one solution for the equation $f(x) = 0$ in the interval (a, b) .

Types of discontinuity :

- **Discontinuity of first kind (or) Removable discontinuity :**

If $\lim_{x \rightarrow a} f(x)$ exists but is not equal to $f(a)$

(or) $f(a)$ not defined then the f is said to have a removable discontinuity at $x = a$. It is also called discontinuity of the 1st kind. In this case we can redefine the function by making

$\lim_{x \rightarrow a} f(x) = f(a)$ and make it continuous at $x = a$.

Removable discontinuities are of two types

- 1) Missing point discontinuity
- 2) Isolated point discontinuity

- **Missing point discontinuity :**

$\lim_{x \rightarrow a} f(x)$ exists finitely and $f(a)$ is not defined.

Example: $f(x) = \frac{(2-x)(x^2-8)}{(2-x)}$ has a missing point discontinuity at $x = 2$.

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- **Isolated point discontinuity :**

$\lim_{x \rightarrow a} f(x)$ exists finitely and $f(a)$ is defined

but $\lim_{x \rightarrow a} f(x) \neq f(a)$.

Example: $f(x) = [x] + [-x]$ has isolated point discontinuities at all integers.

- **Discontinuity of second kind (or) irremovable discontinuity :**

A function $f(x)$ is said to have a discontinuity of the second kind at $x = a$ if $\lim_{x \rightarrow a} f(x)$ does not exist.

Irremovable discontinuities are of three types

- 1) Finite discontinuity (or) jump discontinuity
- 2) Infinite discontinuity
- 3) Oscillatory discontinuity

- **Finite Discontinuity:**

$\lim_{x \rightarrow a^+} f(x), \lim_{x \rightarrow a^-} f(x)$ are both finite and are not equal.

Example: $f(x) = \frac{1}{1+2^x}$ at $x = 0$

- **Infinite Discontinuity :**

If at least one of the limits $\lim_{x \rightarrow a^+} f(x)$ and

$\lim_{x \rightarrow a^-} f(x)$ be $\pm\infty$, then $f(x)$ has infinite discontinuity at $x = a$.

Example: $f(x) = \frac{\cos x}{x}$ at $x = 0$.

- **Oscillatory Discontinuity:**

The Limit oscillates between two finite quantities

Example: $f(x) = \sin \frac{1}{x}$ at $x = 0$.

- In case of discontinuity of the second kind, the absolute difference between the value of the RHL at $x = a$ and LHL at $x = a$ is called the Jump of Discontinuity. A function having a finite number of jumps in a given interval I is called a Piece wise continuous or Sectionally continuous function in this interval.

W.E-4: The jump of discontinuity of the function

$$f(x) = \frac{|2x-3|}{2x-3} \text{ is}$$

Sol: $f\left(\frac{3}{2}+\right) = 1, f\left(\frac{3}{2}-\right) = -1$

$\therefore$ Jump of discontinuity = 2

- All polynomials, Trigonometrical functions, exponential and Logarithmic functions are continuous in their respective domains.

Intermediate value theorem :

- Suppose $f(x)$ is continuous on a interval I, and a, b are any two points of I. If y_0 is a number between $f(a)$ and $f(b)$, then there exists a number c between a and b such that $f(c) = y_0$.

Single Point Continuity:

- Functions which are continuous only at one point are said to exhibit single point continuity behaviour.

Example 1: $f(x) = \begin{cases} x, & \text{if } x \in Q \\ -x, & \text{if } x \notin Q \end{cases}$ is continuous only at $x = 0$.

Example 2: $f(x) = \begin{cases} x, & \text{if } x \in Q \\ 1-x, & \text{if } x \notin Q \end{cases}$ is continuous only at $x = 1/2$

W.E-5: If $y = \frac{1}{t^2+t-2}$ where $t = \frac{1}{x-1}$, then the number of points of discontinuous of $y = f(x), x \in R$ is

Sol: $t = \frac{1}{x-1}$ is discontinuous at $x = 1$. Also

$y = \frac{1}{t^2+t-2}$ is discontinuous at $t = -2$ and $t = 1$

When $t = -2, \frac{1}{x-1} = -2 \Rightarrow x = \frac{1}{2},$

When $t = 1, \frac{1}{x-1} = 1 \Rightarrow x = 2,$

So, $y = f(x)$ is discontinuous at three points, $x = 1, \frac{1}{2}, 2$



C.U.Q.

- $|x|$ is continuous at
 - everywhere of x
 - $x = 0$ only
 - $(1, \infty)$ only
 - $(-\infty, 1)$ only
- $\text{Sgn}(x)$ is discontinuous at
 - everywhere of x
 - $x = 0$
 - $(1, \infty)$
 - $(-\infty, 1)$
- $[x]$ is continuous at
 - $R - Z$
 - Z
 - N
 - W
- $\{x\}$ is discontinuous at (here $\{ \}$ is a fractional part function)
 - every where of x
 - Z
 - N
 - $R - Z$

C.U.Q - KEY

- 1) 1 2) 2 3) 1 4) 4

C.U.Q - HINTS

- 1) By using the graph

2) function = $\begin{cases} |x| & \text{at } x \neq 0 \\ x & \text{at } x = 0 \end{cases}$

- 3) By using the graph

- 4) By using the graph



LEVEL-I(C.W.)

CONTINUITY AT A POINT

1. The function

$$f(x) = \begin{cases} \frac{\cos 3x - \cos 4x}{x^2}, & \text{for } x \neq 0 \\ \frac{7}{2}, & \text{for } x = 0 \end{cases}$$

at $x = 0$ is

(EAM - 2007)

- continuous
- discontinuous
- left continuous
- right continuous

2. The function defined by

$$f(x) = \begin{cases} x \cdot \sin \frac{1}{x} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases} \quad \text{at } x = 0 \text{ is}$$

- continuous
- right continuous
- left continuous
- can not be determined

3. If the function $f(x) = \frac{e^{x^2} - \cos x}{x^2}$ for $x \neq 0$ is continuous at $x=0$ then $f(0) =$
 1) $1/2$ 2) $3/2$ 3) 2 4) $1/3$
4. The value of $f(0)$ so that the function $f(x) = \frac{\log\left(1+\frac{x}{a}\right) - \log\left(1-\frac{x}{b}\right)}{x}, (x \neq 0)$ is continuous at $x=0$ is
 1) $\frac{a+b}{ab}$ 2) $\frac{a-b}{ab}$ 3) $\frac{ab}{a+b}$ 4) $\frac{ab}{a-b}$
5. $f(x) = x\left[3 - \log\left(\frac{\sin x}{x}\right)\right] - 2$ to be continuous at $x=0$, then $f(0) =$
 1) 0 2) 2 3) -2 4) 3
6. If $f(x) = x^{\frac{1}{x-1}}$ for $x \neq 1$ and f is continuous at $x=1$ then $f(1) =$
 1) e 2) e^{-1} 3) e^{-2} 4) e^2
7. If $f(x) = \begin{cases} \frac{1-\sqrt{2}\sin x}{\pi-4x}, & \text{if } x \neq \frac{\pi}{4} \\ a, & \text{if } x = \frac{\pi}{4} \end{cases}$ is continuous at $x = \frac{\pi}{4}$ then $a =$
 1) 4 2) 2 3) 1 4) $1/4$
8. The discontinuous points of $f(x) = \frac{1}{\log|x|}$ are
 1) $0, \pm 2$ 2) $1, \pm 2$ 3) $0, \pm 1$ 4) $0, \pm 3$
9. Let $f: R \rightarrow R$ be defined by $f(x) = \begin{cases} \alpha + \frac{\sin[x]}{x}, & \text{if } x > 0 \\ 2, & \text{if } x = 0 \\ \beta + \left[\frac{\sin x - x}{x^3}\right], & \text{if } x < 0 \end{cases}$ where $[x]$ denotes the integral part of x . If f is continuous at $x=0$, then $\beta - \alpha =$ [EAM-12]
 1) -1 2) 1 3) 0 4) 2

10. If $[x]$ denotes a greatest integer not exceeding x and if the function f defined

$$\text{by } f(x) = \begin{cases} \frac{a+2\cos x}{x^2}, & \text{if } x < 0 \\ b \tan \frac{a}{[x+4]}, & \text{if } x \geq 0 \end{cases}$$

is continuous at $x=0$, then the ordered pair (a, b) is [EAM - 2011]

- 1) $(-2, 1)$ 2) $(-2, -1)$
 3) $(-1, \sqrt{3})$ 4) $(-2, -\sqrt{3})$

CONTINUITY IN AN INTERVAL

11. If $f: R \rightarrow R$ is defined by

$$f(x) = \begin{cases} \frac{x+2}{x^2+3x+2} & \text{if } x \in R - \{-1, -2\} \\ -1 & \text{if } x = -2 \\ 0 & \text{if } x = -1 \end{cases}$$

then f is continuous on the set [EAM-05]

- 1) R 2) $R - \{-2\}$
 3) $R - \{-1\}$ 4) $R - \{-1, -2\}$

12. Let f be a non-zero real valued continuous function satisfying $f(x+y) = f(x) \cdot f(y)$ for all $x, y \in R$. If $f(2)=9$, then $f(6)=$ (EAM-13)

- 1) 3^2 2) 3^6 3) 3^4 4) 3^3

13. If $f: [-2, 2] \rightarrow R$ is defined by

$$f(x) = \begin{cases} \frac{\sqrt{1+cx} - \sqrt{1-cx}}{x}, & \text{for } -2 \leq x < 0 \\ \frac{x+3}{x+1}, & \text{for } 0 \leq x \leq 2 \end{cases}$$

is continuous on $[-2, 2]$ then $c =$ (EAM-14)

- 1) 3 2) $\frac{3}{2}$ 3) $\frac{3}{\sqrt{2}}$ 4) $\frac{2}{\sqrt{3}}$

14. $f(x) = \frac{p+q^{\frac{1}{x}}}{r+s^{\frac{1}{x}}}, s > 1, q > 1, r \neq 0, f(0)=1$ is

left continuous at $x=0$ then

- 1) $p=0$ 2) $p=r$ 3) $p=q$ 4) $p \neq q$

15. If $f(x) = \begin{cases} \sin x, & \text{if } x \text{ is rational} \\ \cos x, & \text{if } x \text{ is irrational} \end{cases}$
then the function is

- 1) discontinuous at $x = n\pi + \frac{\pi}{4}$
- 2) continuous at $x = n\pi + \frac{\pi}{4}$
- 3) discontinuous at all x
- 4) continuous at all x

16. $f(x) = |x| + |x-1|$ is continuous at

- 1) 0 only
- 2) 0, 1 only
- 3) every where
- 4) no where

LEVEL-I (C.W)-KEY

- 1) 1 2) 1 3) 2 4) 1 5) 3 6) 1
7) 4 8) 3 9) 2 10) 2 11) 3 12) 2
13) 1 14) 2 15) 2 16) 3

LEVEL-I (C.W)-HINTS

1. $\lim_{x \rightarrow 0} \frac{\cos ax - \cos bx}{x^2} = \frac{b^2 - a^2}{2}$ and $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
2. $f(x)$ is continuous at $x = 0$
3. Applying L-Hospital rule
$$f(0) = \lim_{x \rightarrow 0} \frac{e^{x^2} \cdot 2x + \sin x}{2x} = \frac{3}{2}$$
4. Use L-Hospital rule
5. $f(0) = -2$
6. $f(1) = e^{\lim_{x \rightarrow 1} \frac{1}{x-1} (x-1)}$
7. Use L-Hospital rule
8. $f(x)$ is discontinuous at $x = 0, -1, 1$
9. $\alpha + 0 = 2 = \beta - 1$
 $\alpha = \beta - 1 \Rightarrow \beta - \alpha = 1$
10. $f(0) = b$
to define $f(0^-)$, $a = -2 \Rightarrow b = -1$.
11. $f(x)$ is not continuous at -1
12. $f(x) = 3^x$
13. $f(0+) = f(0-)$
14. Put $x = -h$ as $h \rightarrow 0$
15. $\sin x = \cos x \Rightarrow x = n\pi + \frac{\pi}{4}$
16. Everywhere



LEVEL-I - (H.W)

CONTINUITY AT A POINT

1. The function $f(x) = \frac{x \tan 2x}{\sin 3x \cdot \sin 5x}$, for $x \neq 0$, is continuous at $x = 0$, then $f(0)$

- 1) $\frac{2}{13}$
- 2) $\frac{2}{17}$
- 3) $\frac{2}{11}$
- 4) $\frac{2}{15}$

2. If $f(x) = \frac{1 - \cos ax}{1 - \cos bx}$, for $x \neq 0$, is continuous at $x = 0$ then $f(0) =$

- 1) $\frac{a^2}{2}$
- 2) $\frac{a}{b^2}$
- 3) $\frac{a}{b}$
- 4) $\frac{a^2}{b^2}$

3. If the function $f(x) = \begin{cases} \frac{\sin^2 ax}{x^2}, & \text{for } x \neq 0 \\ 1, & \text{for } x = 0 \end{cases}$

is continuous at $x = 0$ then $a =$

- 1) ± 1
- 2) 0
- 3) $\pm \frac{1}{2}$
- 4) $\pm \frac{1}{3}$

4. If the function $f(x) = \begin{cases} \frac{2^{x+2} - 16}{4^x - 16}, & \text{for } x \neq 2 \\ A, & \text{for } x = 2 \end{cases}$

is continuous at $x = 2$, then $A =$

- 1) 2
- 2) $1/2$
- 3) $1/4$
- 4) 0

5. If $f(x) = \begin{cases} \frac{(e^{kx} - 1) \cdot \sin kx}{x^2}, & \text{for } x \neq 0 \\ 4, & \text{for } x = 0 \end{cases}$

is continuous at $x = 0$ then $k =$

- 1) ± 1
- 2) ± 2
- 3) 0
- 4) ± 3

6. Let $f(x) = \frac{x + x^2 + \dots + x^n - n}{x - 1}$, $x \neq 1$, the value of $f(1)$ so that f is continuous at $x = 1$ is

- 1) n
- 2) $\frac{n+1}{2}$
- 3) $\frac{n(n+1)}{2}$
- 4) $\frac{n(n-1)}{2}$

CONTINUITY

7. The set of points of discontinuity of the

$$\text{function } f(x) = \frac{1}{x^2 + x + 1}$$

- 1) ϕ 2) $\mathbb{R}$ 3) $\{0\}$ 4) $\mathbb{R}^-$

8. If $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{1+3x^2 - \cos 2x}{x^2}, & \text{for } x \neq 0 \\ k, & \text{for } x = 0 \end{cases}$$

is continuous at $x=0$, then $k =$ [EAM - 2010]

- 1) 1 2) 5 3) 6 4) 0

9. If $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} \frac{2 \sin x - \sin 2x}{2x \cos x}, & \text{if } x \neq 0 \\ a, & \text{if } x = 0 \end{cases} \quad \text{then the}$$

value of a so that f is continuous at $x=0$ is

[EAM - 2009]

- 1) 2 2) 1 3) -1 4) 0

10. If $f(x) = \begin{cases} \frac{\sqrt{1+kx} - \sqrt{1-kx}}{x}, & \text{for } -1 \leq x < 0 \\ 2x^2 + 3x - 2, & \text{for } 0 \leq x \leq 1 \end{cases}$

is continuous at $x=0$ then $k =$

- 1) -4 2) -3 3) -2 4) -1

CONTINUITY IN AN INTERVAL

11. The interval on which $f(x) = \sqrt{1-x^2}$ is continuous in

- 1) $(0, \infty)$ 2) $(1, \infty)$ 3) $[-1, 1]$ 4) $(-\infty, -1)$

12. The function $f(x) = \begin{cases} 0, & \text{if } x \text{ is irrational} \\ 1, & \text{if } x \text{ is rational} \end{cases}$ is

- 1) continuous at $x=1$
2) discontinuous only at 0
3) discontinuous only at 0,1
4) discontinuous everywhere

CONTINUITY OF SPECIAL TYPE OF FUNCTIONS

13. If $f(x)$ is a continuous function, then

$$\lim_{x \rightarrow 0} f(x) \frac{|x|}{x} \text{ exist if}$$

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1) $f(x)$ is a polynomial 2) $f(x) = ax^2 + bx + c$

3) $f(x) = ax^2 + bx$ 4) $f(x) = ax + b$

14. The function $f(x) = \frac{|x-3|}{x-3}$ at $x=3$, is

- 1) Left continuous 2) Right continuous
3) Continuous 4) Discontinuous

LEVEL-I (H.W)-KEY

- 1) 4 2) 4 3) 1 4) 2 5) 2 6) 3
7) 1 8) 2 9) 4 10) 3 11) 3 12) 4
13) 3 14) 4

LEVEL-I (H.W)-HINTS

$$1. f(0) = k = \lim_{x \rightarrow 0} \frac{x \tan 2x}{\sin 3x \sin 5x} = \frac{2}{15}$$

$$2. f(0) = \frac{a^2}{b^2}$$

$$3. \lim_{x \rightarrow 0} \frac{\sin^2 ax}{x^2} = 1$$

$$4. A = \lim_{x \rightarrow 2} f(x)$$

$$= \lim_{x \rightarrow 2} \frac{2^x \cdot 2^2 - 16}{4^x - 16} = \lim_{x \rightarrow 2} \frac{2^2 \cdot 2^x \log 2}{4^x \cdot \log 4}$$

$$= \frac{4^2 \cdot \log 2}{4^2 \cdot \log 4} = \log_4 2 = \log_{2^2} 2^1 = \frac{1}{2}$$

$$5. f(0) = \lim_{x \rightarrow 0} \left(\frac{e^{kx} - 1}{x} \right) \cdot \frac{\sin kx}{x} \Rightarrow 4 = k^2$$

$$6. f(1) = \lim_{x \rightarrow 1} \frac{x + x^2 + \dots + x^n - n}{x - 1} \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{x \rightarrow 1} \frac{1 + 2x + \dots + nx^{n-1}}{1} = 1 + 2 + \dots + n$$

$$= \frac{n(n+1)}{2}$$

7. $f(x)$ is defined for any $x \in \mathbb{R}$

8. Use L-Hospital's Rule

9. Simplify

10. $f(x)$ is continuous at 0

11. $1 - x^2 \geq 0; x \in [-1, 1]$

12. If 'a' is any real number, then every neighbourhood of a contains infinitely many rationals as well as irrationals. So $f(x)$ assumes values 0 and 1 in every neighbourhood of a and hence $\lim_{x \rightarrow a} f(x)$ does not exist.
So, f is not continuous at any real number.

13. $\lim_{x \rightarrow 0^+} \frac{|x|}{x} = 1$ and $\lim_{x \rightarrow 0^-} \frac{|x|}{x} = -1$

Also $f(x)$ is continuous

$\therefore$ Given limit can exist only if

$$\lim_{x \rightarrow 0} f(x) = 0$$

$\therefore f(x) = ax^2 + bx$ is the only choice

14. $f(x)$ is not defined at $x = 3$



LEVEL-II - (C.W)

CONTINUITY AT A POINT

1. Let

$$f(x) = \begin{cases} \frac{(e^x - 1)^{2n}}{\sin^n(x/a) (\log(1 + (x/a)))^n}, & \text{for } x \neq 0 \\ 16^n, & \text{for } x = 0 \end{cases}$$

and f is a continuous at $x=0$, then the value of a is

- 1) 16 2) 2 3) 8 4) 4

2. If $f(x) = \begin{cases} x^\alpha \cos(1/x), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ is continuous

at $x = 0$ then

- 1) $\alpha < 0$ 2) $\alpha > 0$ 3) $\alpha = 0$ 4) $\alpha \geq 0$

3. Let $f(x) = \begin{cases} \frac{x(1 + a \cos x) - b \sin x}{x^3}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$

The values of a and b so that f is a continuous function at $x=0$, are

- 1) $5/2, 3/2$ 2) $5/2, -3/2$
3) $-5/2, -3/2$ 4) $-\frac{5}{2}, \frac{3}{2}$

4. If $f(x) = \begin{cases} a^2 \cos^2 x + b^2 \sin^2 x, & x \leq 0 \\ e^{ax+b}, & x > 0 \end{cases}$

$f(x)$ is continuous at $x = 0$ then

1) $2 \log|a| = b$

3) $\log a = 2 \log|b|$

2) $2 \log|b| = c$

4) $a = b$

5. $f(x) = \begin{cases} \left(\frac{3}{x^2}\right) \sin 2x^2, & \text{if } x < 0 \\ \frac{x^2 + 2x + c}{1 - 3x^2}, & \text{if } x \geq 0, x \neq \frac{1}{\sqrt{3}} \\ 0, & \text{if } x = \frac{1}{\sqrt{3}} \end{cases}$

then in order that f be continuous at $x = 0$, the value of c is

- 1) 2 2) 4 3) 6 4) 8

CONTINUITY IN AN INTERVAL

6. If $f(x) = \begin{cases} -1, & \text{if } x < 0 \\ 0, & \text{if } x = 0 \\ 1, & \text{if } x > 0 \end{cases}$

and $g(x) = \sin x + \cos x$, then points of discontinuity of $(f \circ g)(x)$ in $(0, 2\pi)$ are

- 1) $\frac{\pi}{4}, \frac{5\pi}{4}$ 2) $\frac{\pi}{4}, \frac{3\pi}{4}$ 3) $\frac{\pi}{4}, \frac{7\pi}{4}$ 4) $\frac{3\pi}{4}, \frac{7\pi}{4}$

CONTINUITY OF SPECIAL TYPE OF FUNCTIONS

7. If $f(x)$ be a continuous function for all real values of x and satisfies

$$x^2 + x(f(x) - 2) + 2\sqrt{3} - 3 - \sqrt{3}f(x) = 0$$

$\forall x \in R$ then the value of $f(\sqrt{3})$ is

- 1) $\sqrt{3}$ 2) $2(\sqrt{3} - 1)$
3) $2\sqrt{3} - 1$ 4) $2(1 - \sqrt{3})$

8. Let f be a continuous function on R such that

$$f\left(\frac{1}{4n}\right) = \sin(e^n) e^{-n^2} + \frac{n^2}{n^2 + 1}. \text{ Then the}$$

value of $f(0)$ is

- 1) 1 2) $1/2$ 3) 0 4) 2

9. The function $f(x) = a[x+1] + b[x-1]$, ($a \neq 0, b \neq 0$) where $[x]$ is the greatest integer function is continuous at $x = 1$ if

- 1) $a = 2b$ 2) $a = b$
3) $a + b = 0$ 4) $a + 2b = 0$

CONTINUITY

10. Let $f(x) = [2x^3 - 6]$ when $[x]$ is greatest integer less than or equal to x then the number of points in $(1, 2)$ where f is discontinuous is

1) 5 2) 7 3) 13 4) 12

11. If $f(x) = \begin{cases} \frac{\sin[x]}{[x]}, & \text{for } [x] \neq 0 \\ 0, & \text{for } [x] = 0 \end{cases}$ is

1) continuous at $x = 0$
 2) discontinuous at $x = 0$
 3) L.H.L = 0 4) R.H.L = 1

12. If $x + 2|y| = 3y$, where $y = f(x)$, then $f(x)$ is

1) continuous everywhere
 2) differentiable everywhere
 3) discontinuous at $x = 0$
 4) Not differentiable at anywhere

13. $f(x) = \min\{x, x^2\}$, $\forall x \in \mathbb{R}$ then $f(x)$ is

1) discontinuous at 0 2) discontinuous at 1
 3) continuous on $\mathbb{R}$ 4) continuous at 0, 1

LEVEL-II (C.W)-KEY

1) 4 2) 2 3) 3 4) 1 5) 3 6) 4
 7) 4 8) 1 9) 3 10) 3 11) 2 12) 1
 13) 3

LEVEL-II (C.W)-HINTS

1. $\lim_{x \rightarrow 0} f(x) =$

$$\lim_{x \rightarrow 0} \left(\frac{e^x - 1}{x} \right)^{2n} \frac{(x/a)^n}{\sin^n(x/a)} \times \frac{(x/a)^n}{(\log(1+(x/a)))^n} \cdot a^{2n}$$

$$= a^{2n}$$

since, $f(0) = \lim_{x \rightarrow 0} f(x)$ so

$$a^{2n} = 16^n = 4^{2n} \text{ thus } a = 4.$$

2. Continuous when $a > 0$
 3. Note that f is continuous everywhere except possibly at $x = 0$. For f to be continuous at $x = 0$,

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$$1 = f(0) = \lim_{x \rightarrow 0} \frac{x(1 + a \cos x) - b \sin x}{x^3}, \left(\frac{0}{0} \text{ form} \right)$$

$$= \lim_{x \rightarrow 0} \frac{(1 + a \cos x) - xa \sin x - b \cos x}{3x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 + (a - b) \cos x - xa \sin x}{3x^2}$$

This limit will exist only if $1 + (a - b) = 0$.

$$= \lim_{x \rightarrow 0} \frac{-(a - b) \sin x - ax \cos x - a \sin x}{6x}$$

$$= \frac{1}{6} \lim_{x \rightarrow 0} \left[(b - 2a) \frac{\sin x}{x} - a \cos x \right] = \frac{1}{6} [b - 3a]$$

$$\Rightarrow b - 3a = 6$$

Solving $a - b = -1$ and $b - 3a = 6$, we get $b = -3/2$ and $a = -5/2$.

4. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$

5. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{3}{x^2} \sin 2x^2 = 6 \lim_{x \rightarrow 0} \frac{\sin 2x^2}{2x^2} = 6$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{x^2 + 2x + c}{1 - 3x^2} = \frac{c}{1} = c$$

Hence for f to be continuous $c = 6$.

$$6. f(g(x)) = \begin{cases} 1, & \text{if } 0 < x < \frac{3\pi}{4} \text{ or } \frac{7\pi}{4} < x < 2\pi \\ 0, & \text{if } x = \frac{3\pi}{4}, \frac{7\pi}{4} \\ -1, & \text{if } \frac{3\pi}{4} < x < \frac{7\pi}{4} \end{cases}$$

Clearly $(f \circ g)(x)$ is not continuous at

$$x = \frac{3\pi}{4}, \frac{7\pi}{4}$$

7. $f(x)$ is continuous at $x = \sqrt{3}$

$$\Rightarrow f(\sqrt{3}) = \lim_{x \rightarrow \sqrt{3}} \frac{x^2 - 2x + 2\sqrt{3} - 3}{\sqrt{3} - x}$$

8. $f(0) = \lim_{x \rightarrow 0} f(x)$

$$= \lim_{n \rightarrow \infty} f\left(\frac{1}{4n}\right) = 0 + 1 = 1$$

9. $f(1) = 2a$, $\lim_{x \rightarrow 1^-} f(x) = a - b$ so
 $a + b = 0$ for f to be continuous at
 $x = 1$.
10. $1 < x < 2 \Rightarrow 1 < x^3 < 8$
 $\Rightarrow 2 < 2x^3 < 16 \Rightarrow -4 < 2x^3 - 6 < 10$
 $\therefore [2x^3 - 6]$ is discontinuous at
 $-3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9$

11. $L.H.L = \lim_{x \rightarrow 0^-} \frac{\sin(-1)}{-1} = \sin 1$

$(\because x \rightarrow 0^- \Rightarrow [x] = -1)$

$R.H.L = 0$ $(\because x \rightarrow 0^+ \Rightarrow [x] = 0)$

12. $x + 2|y| = 3y$
 $\Rightarrow x + 2y = 3y, y \geq 0$ and
 $x - 2y = 3y, y < 0$

$$\Rightarrow y = \begin{cases} x, & x \geq 0 \\ \frac{x}{5}, & x < 0 \end{cases}$$

Therefore, y is continuous everywhere but
is not differentiable at $x = 0$.

13. $x = x^2 \Rightarrow x = 0, x = 1$

$$f(x) = \begin{cases} x & \text{for } x < 0 \\ x^2 & \text{for } 0 \leq x \leq 1 \\ x & \text{for } x > 1 \end{cases}$$



LEVEL-II - (H.W)

CONTINUITY AT A POINT

1. If $f(x) = \frac{\sqrt{a^2 - ax + x^2} - \sqrt{a^2 + ax + x^2}}{\sqrt{a+x} - \sqrt{a-x}}$ is

continuous at $x = 0$ then $f(0) =$

1) $\sqrt{a}$ 2) $-\sqrt{a}$ 3) $a\sqrt{a}$ 4) $-a\sqrt{a}$

2. If $f(x) = \begin{cases} \frac{72^x - 9^x - 8^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}}, & \text{if } x \neq 0 \\ K \log 2 \log 3, & \text{if } x = 0 \end{cases}$ is a

continuous function, then K is equal to

1) $\sqrt{2}$ 2) 24 3) $18\sqrt{3}$ 4) $24\sqrt{2}$

CONTINUITY IN AN INTERVAL

3. The value of $f(0)$ so that the function

$$f(x) = \frac{1 - \cos(1 - \cos x)}{x^4} \quad \text{is continuous}$$

everywhere is

1) $\frac{1}{8}$ 2) $\frac{1}{2}$ 3) $\frac{1}{4}$ 4) $\frac{1}{3}$

4. The function $y = 3\sqrt{x} - |x - 1|$ is continuous at

1) $x < 0$ 2) $x \geq 1$ 3) $0 \leq x \leq 1$ 4) $x \geq 0$

5. If the function $f(x) = \begin{cases} \frac{1 - \cos 4x}{x^2}, & \text{if } x < 0 \\ a, & \text{if } x = 0 \\ \frac{\sqrt{x}}{\sqrt{16 + \sqrt{x}} - 4}, & \text{if } x > 0 \end{cases}$ is

continuous at $x = 0$ then $a =$

1) 8 2) $\frac{1}{8}$ 3) -8 4) 0

6. $f(x) = \begin{cases} \frac{(x + bx^2)^{1/2} - x^{1/2}}{bx^{3/2}}, & \text{if } x > 0 \\ c, & \text{if } x = 0 \\ \frac{\sin(a+1)x + \sin x}{x}, & \text{if } x < 0 \end{cases}$

is continuous at $x = 0$ then

1) $a = \frac{-3}{2}, b = 0, c = \frac{1}{2}$ 2) $a = \frac{-3}{2}, b \neq 0, c = \frac{1}{2}$

3) $a = \frac{3}{2}, b \neq 0, c = \frac{1}{2}$ 4) $a = \frac{3}{2}, b \neq 0, c = -\frac{1}{2}$

7. If $f(x) = \begin{cases} (1 + |\sin x|)^{\frac{a}{|\sin x|}}, & \text{if } -\frac{\pi}{6} < x < 0 \\ b, & \text{if } x = 0 \\ e^{\frac{\tan 2x}{\tan 3x}}, & \text{if } 0 < x < \frac{\pi}{6} \end{cases}$ is

continuous at $x = 0$ then

1) $a = e^{2/3}, b = 2/3$ 2) $a = 2/3, b = e^{2/3}$

3) $a = 1/3, b = e^{1/3}$ 4) $a = e^{1/3}, b = e^{1/3}$

CONTINUITY OF SPECIAL TYPE OF FUNCTIONS

8. Let $f(x) = g(x) \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}}$, where g is a continuous function then $\lim_{x \rightarrow 0} f(x)$ exist if

- 1) $g(x) = x + 2$ 2) $g(x) = x^2 + 4$
- 3) $g(x) = xh(x)$, $h(x)$ is a polynomial
- 4) $g(x)$ is a constant function

9. The function $f(x) = [\cos x]$ is

- 1) continuous at $x = \frac{\pi}{2}$
- 2) discontinuous at $x = \frac{\pi}{2}$
- 3) L.H.L = -1 at $x = \frac{\pi}{2}$
- 4) R.H.L = 1 at $x = \frac{\pi}{2}$

10. The function $f(x) = \frac{1}{x^2 - 3|x| + 2}$ is discontinuous at the points

- 1) $x = 1, 2$ 2) $x = \pm 1, \pm 2$
- 3) R 4) $R - \{1, 2\}$

11. Let a function f be defined by $f(x) = \frac{x - |x|}{x}$

for $x \neq 0$ and $f(0) = 2$, then f is

- 1) continuous no where
- 2) continuous every where
- 3) continuous for all x except $x = 1$
- 4) continuous for all x except $x = 0$

12. If $f(x) = \frac{1}{1-x}$ then the points of discontinuity of $(f \circ f \circ f)(x)$ is

- 1) $\{0, 1\}$ 2) $\{0, \pm 1\}$ 3) $\{1\}$ 4) $\{\pm 1\}$

LEVEL-II (H.W)-KEY

- 1) 2 2) 4 3) 1 4) 4 5) 1 6) 2
- 7) 2 8) 3 9) 2 10) 2 11) 4 12) 1

LEVEL-II (H.W)-HINTS

1. $f(0) = \lim_{x \rightarrow 0} f(x)$ and rationalise or use L.H. rule
2. $K \log 2 \log 3 = f(0)$

$$\begin{aligned} &= \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{72^x - 9^x - 8^x + 1}{\sqrt{2} - \sqrt{1 + \cos x}} \\ &= \lim_{x \rightarrow 0} \frac{(9^x - 1)(8^x - 1)}{x^2} \cdot \frac{x^2}{\sqrt{2}(1 - \cos x/2)} \\ &= \lim_{x \rightarrow 0} \frac{9^x - 1}{x} \cdot \frac{8^x - 1}{x} \cdot \frac{16(x/4)^2}{\sqrt{2} \cdot 2 \sin^2 x/4} \\ &= \frac{16}{2\sqrt{2}} \log 9 \log 8 \\ &= \frac{8}{\sqrt{2}} 6 \log 3 \log 2 = 24\sqrt{2} \log 3 \log 2 \end{aligned}$$

Thus $K = 24\sqrt{2}$.

$$\begin{aligned} 3. \quad &\lim_{x \rightarrow 0} \frac{1 - \cos(1 - \cos x)}{(1 - \cos x)^2} \cdot \frac{(1 - \cos x)^2}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{1 - \cos(1 - \cos x)}{(1 - \cos x)^2} \left(\frac{(1 - \cos x)}{x^2} \right)^2 \\ &= \frac{1}{2} \times \frac{1}{4} \left(\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \right) (\text{formula}) \\ &= \frac{1}{8} \end{aligned}$$

$$\begin{aligned} 4. \quad &\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 3\sqrt{x} - |x - 1| \\ &= 0 - |-1| = -1 \end{aligned}$$

Since $\lim_{x \rightarrow 0^+} f(x) = f(0)$

$\therefore f(x)$ is continuous for $x \geq 0$

$$5. \quad a = \lim_{x \rightarrow 0^-} \frac{1 - \cos 4x}{x^2} = 8$$

$$6. \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = f(0)$$

$$7. \quad \lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^-} f(x) = f(0)$$

$$\begin{aligned} 8. \quad &\lim_{x \rightarrow 0^+} \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} = \lim_{x \rightarrow 0^+} \frac{1 - e^{-2/x}}{1 + e^{-2/x}} = 1 \\ &\lim_{x \rightarrow 0^-} \frac{e^{1/x} - e^{-1/x}}{e^{1/x} + e^{-1/x}} = \lim_{x \rightarrow 0^-} \frac{e^{2/x} - 1}{e^{2/x} + 1} = -1 \end{aligned}$$

Hence $\lim_{x \rightarrow 0} f(x)$ exists if $g(x) = xh(x)$

where $h(x)$ is a continuous function. If

$g(x) = a(a \neq 0)$ then

$\lim_{x \rightarrow 0^+} f(x) = a, \lim_{x \rightarrow 0^-} f(x) = -a$. Thus

$\lim_{x \rightarrow 0} f(x)$ does not exist if $f(x) = x + 2$

or $x^2 + 4$ or is a constant function.

9. $L.H.L = \lim_{x \rightarrow \frac{\pi}{2}} [\cos x] = 0$

$R.H.L = \lim_{x \rightarrow \frac{\pi}{2}^+} [\cos x] = -1$

10. $f(x)$ is discontinuous when

$$x^2 - 3|x| + 2 = 0 \Rightarrow |x|^2 - 3|x| + 2 = 0 \Rightarrow |x| = 1, 2$$

11. $f(x) = \begin{cases} 2, & \text{if } x < 0 \\ 0, & \text{if } x > 0 \\ 2, & \text{if } x = 0 \end{cases}$

Thus $\lim_{x \rightarrow 0^-} f(x) = 2 \neq 0 = \lim_{x \rightarrow 0^+} f(x)$.

Hence, f is continuous every where except at $x = 0$.

12. $f(x) = \frac{1}{1-x}$

$$(fof)(x) = \frac{1}{1 - \frac{1}{1-x}} = \frac{1-x}{-x} = \frac{x-1}{x}$$

$$(fofof)(x) = \frac{1}{1 - \frac{x-1}{x}} = x$$

$(fofof)(x)$ is discontinuous at $x = 0, x = 1$.



LEVEL-III

CONTINUITY AT A POINT

1. If $f(x) = \begin{cases} \frac{|x+2|}{\tan^{-1}(x+2)}, & \text{if } x \neq -2 \\ 2, & \text{if } x = -2 \end{cases}$ then, $f(x)$ is

- 1) continuous at $x = -2$
- 2) not continuous at $x = -2$
- 3) differentiable at $x = -2$
- 4) continuous but not differentiable at $x = -2$

2. Let $f(x) = \frac{(256+ax)^{1/8} - 2}{(32+bx)^{1/5} - 2}$. If f is continuous at $x = 0$, then the value of a/b is

- 1) $\frac{8}{5}f(0)$
- 2) $\frac{32}{5}f(0)$
- 3) $\frac{64}{5}f(0)$
- 4) $\frac{16}{5}f(0)$

3. If $f^n(x)$ be continuous at

$x = 0, f(0) \neq 0, f^1(0) \neq 0$ and

$$\lim_{x \rightarrow 0} \frac{2f(x) - 3af(2x) + bf(8x)}{\sin^2 x},$$

exists then the values of a and b are

- 1) $a = \frac{7}{9}, b = \frac{1}{3}$
- 2) $a = 1, b = 1$
- 3) $a = b = -1$
- 4) $a = -1, b = 1$

4. The values of a and b if f is continuous at $x = 0$, where

$$f(x) = \begin{cases} \left(1 + \frac{ax+bx^3}{x^2}\right)^{1/x}, & \text{if } x > 0 \\ 3, & \text{if } x = 0 \end{cases}$$

- 1) $a = 0, b = \log 3$
- 2) $a = 1, b = \log 2$
- 3) $a = 2, b = \log 3$
- 4) $a = 0, b = \log 2$

5. Let $f(x) = \frac{\log(1+x+x^2) + \log(1-x+x^2)}{\sec x - \cos x}$,

$x \neq 0$. Then the value of $f(0)$ so that f is continuous at $x = 0$ is

- 1) 1
- 2) 0
- 3) 2
- 4) -1

6. If $f(x) = \begin{cases} \frac{\sin(\cos x) - \cos x}{(\pi - 2x)^3}, & \text{if } x \neq \frac{\pi}{2} \\ k, & \text{if } x = \frac{\pi}{2} \end{cases}$

is continuous at $x = \frac{\pi}{2}$, then $k =$

- 1) 0
- 2) $-\frac{1}{6}$
- 3) $-\frac{1}{24}$
- 4) $-\frac{1}{48}$

7. If $f(x) = \lim_{n \rightarrow \infty} \frac{\log(2+x) - x^{2n} \sin x}{1+x^{2n}}$ then $f(x)$ is discontinuous at

- 1) $x = 1$ only
- 2) $x = -1$ only
- 3) $x = -1, 1$ only
- 4) no point

8. Let $f: R \rightarrow R$ be given by

$$f(x) = \begin{cases} 5x, & \text{if } x \in Q \\ x^2 + 6, & \text{if } x \in R - Q \end{cases} \text{ then}$$

- 1) f is continuous at $x = 2$ and $x = 3$
- 2) f is discontinuous at $x = 2$ and $x = 3$
- 3) f is continuous at $x = 2$ but not at $x = 3$
- 4) f is continuous at $x = 3$ but not at $x = 2$

CONTINUITY IN AN INTERVAL

9. If $f(x) = \begin{cases} \frac{A+3\cos x}{x^2}, & \text{if } x < 0 \\ B \tan\left(\frac{\pi}{[x+3]}\right), & \text{if } x \geq 0 \end{cases}$ Where $[.]$

represents the greatest integer function, is continuous at $x = 0$ Then

- 1) $A = -3, B = -\sqrt{3}$
- 2) $A = 3, B = -\frac{\sqrt{3}}{2}$
- 3) $A = -3, B = -\frac{\sqrt{3}}{2}$
- 4) $A = -\frac{\sqrt{3}}{2}, B = -3$

10. If $f(x) = \begin{cases} \frac{1-\sin^3 x}{3\cos^2 x}, & \text{if } x < \frac{\pi}{2} \\ a, & \text{if } x = \frac{\pi}{2} \\ \frac{b(1-\sin x)}{(\pi-2x)^2}, & \text{if } x > \frac{\pi}{2} \end{cases}$,

so that $f(x)$ is continuous at $x = \frac{\pi}{2}$, then

- 1) $a = 1/2$
- 2) $b = 5$
- 3) $a = 1$
- 4) $b = -4$

11. If $f(x) = \begin{cases} \frac{a|x^2-15x+56|}{x-8}, & \text{if } x > 9 \\ b, & \text{if } x = 9 \\ \frac{x-[x]}{x-8}, & \text{if } x < 9 \end{cases}$,

where $[.]$ denotes greatest integer function and the function is continuous then

- 1) $a = \frac{1}{2}, b = 1$
- 2) $a = 0, b = 1$
- 3) $a = \frac{-1}{2}, b = 1$
- 4) $a = \frac{-1}{2}, b = -1$

12. If $f(x) = \begin{cases} bx^2 - a, & \text{if } x < -1 \\ ax^2 - bx - 2, & \text{if } x \geq -1 \end{cases}$, If $f^1(x)$ is continuous everywhere. Then the equation whose roots are a and b is

- 1) $x^2 + 3x - 2 = 0$
- 2) $x^2 - 3x + 2 = 0$
- 3) $x^2 + 3x + 2 = 0$
- 4) $x^2 - 5x + 6 = 0$

13. Let $f(x) = \frac{\log(1+x^2)}{x^4 - 26x^2 + 25}$. Then

- 1) f is continuous on $[6, 10]$
- 2) f is continuous on $[-2, 2]$
- 3) f is continuous on $[-6, 6]$
- 4) f is continuous on $[1, 7]$

14. If $[.]$ denotes the greatest integer function then the number of points where

$$f(x) = [x] + \left[x + \frac{1}{3}\right] + \left[x + \frac{2}{3}\right] \text{ is}$$

discontinuous for $x \in (0, 3)$ are

- 1) 2
- 2) 9
- 3) 8
- 4) 10

15. $f(x) = \begin{cases} x[x], & \text{if } 0 \leq x < 3 \\ (x-1)[x], & \text{if } 3 \leq x \leq 4 \end{cases}$ where $[x]$ is the greatest integer function. The function $f(x)$ is

- 1) Continuous and differentiable at $x = 3$
- 2) Continuous at $x = 1, 2, 3$
- 3) Continuous but not differentiable at $x = 3$
- 4) discontinuous at $x = 1, 2, 3$

16. If the function

$$f(x) = \left[\frac{(x-5)^3}{A} \right] \sin(x-5) + a \cos(x-2)$$

where $[.]$ denotes the greatest integer function, is continuous and differentiable in $(7, 9)$, then

- 1) $A \in [8, 64]$
- 2) $A \in (0, 8]$
- 3) $A \in [64, \infty)$
- 4) $A \in [8, 16]$

17. The function

$$f(x) = \begin{cases} x^2 - ax + 3, & \text{if } x \text{ is rational} \\ 2 - x, & \text{if } x \text{ is irrational} \end{cases} \text{ is}$$

continuous at exactly two points then the possible values of ' a ' are

- 1) $(2, \infty)$
- 2) $(-\infty, 3)$
- 3) $(-\infty, -1) \cup (3, \infty)$
- 4) R

18. A point where the function $f(x)$ is not continuous when $f(x) = [\sin[x]]$ in $(0, 2\pi)$ ($[.]$ denotes greatest integer $\leq x$) is

- 1) $(3, 0)$ 2) $(2, 0)$ 3) $(1, 0)$ 4) $(4, -1)$

19. Let $f(x) = \operatorname{sgn}(x)$ and

$g(x) = x(x^2 - 5x + 6)$. The function $f(g(x))$ is discontinuous at

- 1) infinitely many points 2) exactly one point
3) exactly three points 4) no point

20. Consider the function $f(x) = x - |x - x^2|$, $-1 \leq x \leq 2$. The points of discontinuities of $f(x)$ for $x \in [-1, 2]$ are

- 1) $x = 0, 1$ 2) $x = 1, 2$
3) $x = 0, \frac{1}{2}, 1$ 4) no points of discontinuities

21. If the function $f(x) = \frac{\sqrt{x^2 + kx + 1}}{x^2 - k}$ is

Continuous for every $x \in R$ Then

- 1) $k \in [-2, 0)$ 2) $k \in (0, \infty)$
3) $k \in (-\infty, 0)$ 4) $k \in R$

CONTINUITY OF SPECIAL TYPE OF FUNCTIONS

22. The function defined by $f(x) = (-1)^{[x^3]}$ where $[.]$ denotes the greatest integer function, is

1) Discontinuous for $x = n^{\frac{1}{3}}$ where n is any integer

2) $f\left(\frac{3}{2}\right) = 1$

3) $f^1(x) = 1$ for $-1 < x < 1$

4) Continuous on R

23. If $f(x) = \begin{cases} xe^{-\left(\frac{1}{|x|} + \frac{1}{x}\right)}, & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$, then $f(x)$ is

- 1) continuous for all x , but is not differentiable
2) neither differentiable nor continuous
3) discontinuous everywhere
4) continuous as well as differentiable for all x

24. The function $f(x) = \cos^{-1}(\cos x)$ is

- 1) Continuous at $x = \pi$
2) Discontinuous $x = \pi$
3) Discontinuous at $x = -\pi$
4) $f(x) = x, \forall x \in R$

25. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the largest integer $\leq y$) is discontinuous at

- 1) all integers
2) all integers except 0 and 1
3) all integers except 0
4) all integers except 1

26. If $f(x) = \frac{1}{x^2 - 17x + 66}$ then $f\left(\frac{2}{x-2}\right)$ is discontinuous at x is equal to

- 1) $2, \frac{7}{3}, \frac{25}{11}$ 2) $2, \frac{8}{3}, \frac{24}{11}$
3) $2, \frac{7}{3}, \frac{24}{11}$ 4) $2, 6, 11$

27. The function $f(x) = |2 \operatorname{sgn}(2x)| + 2$ has

- 1) Jump discontinuity
2) Removable discontinuity
3) Infinite discontinuity
4) No discontinuity at $x = 0$

28. $f(x) = \operatorname{Sgn}(x^3 - x)$ is discontinuous at $x =$

- 1) 0 2) 1 3) -1 4) 0, -1, 1

29. If $f(x) = \operatorname{Sgn}(2 \sin x + a)$ is continuous for all x then the possible values of 'a' are

- 1) R 2) $a < -2$ or $a > 2$
3) $(-2, 2)$ 4) $(0, \infty)$

30. If $f: R \rightarrow R$ is a function defined by

$$f(x) = [x] \cos\left(\frac{2x-1}{2}\right)\pi, \text{ where } [x] \text{ denotes}$$

the greatest integer function, then f is

[AIEEE - 2012]

- 1) Continuous for every real x
2) discontinuous only at $x = 0$
3) discontinuous only at non-zero integral values of x .
4) continuous only at $x = 0$.

31. The values of p and q for which the function

$$f(x) = \begin{cases} \frac{\sin(p+1)x + \sin x}{x} & , \text{ if } x < 0 \\ q & , \text{ if } x = 0 \\ \frac{\sqrt{x+x^2} - \sqrt{x}}{\frac{3}{x^2}} & , \text{ if } x > 0 \end{cases}$$

is continuous for all x in R , is [AIEEE - 2011]

- 1) $p = \frac{5}{2}, q = \frac{1}{2}$ 2) $p = -\frac{3}{2}, q = \frac{1}{2}$
 3) $p = \frac{1}{2}, q = \frac{3}{2}$ 4) $p = \frac{1}{2}, q = -\frac{3}{2}$

32. If $f(x) = \lim_{x \rightarrow \infty} (\sin x)^{2n}$, then $f(x)$ is

- 1) discontinuous at $x = \pi/2$
 2) continuous at $x = \pi/2$
 3) discontinuous at $x = -\pi/2$
 4) discontinuous at infinite number of points

LEVEL-III-KEY

- 1) 2 2) 3 3) 1 4) 1 5) 1 6) 4
 7) 3 8) 1 9) 3 10) 1 11) 1 12) 2
 13) 1 14) 3 15) 3 16) 3 17) 3 18) 4
 19) 3 20) 4 21) 1 22) 1 23) 1 24) 1
 25) 4 26) 3 27) 2 28) 4 29) 2 30) 1
 31) 2 32) 4

LEVEL-III-HINTS

1. $\lim_{x \rightarrow -2^-} f(x) = \lim_{x \rightarrow -2^-} \frac{|x+2|}{\tan^{-1}(x+2)}$
 $= \lim_{x \rightarrow -2^-} \frac{-(x+2)}{\tan^{-1}(x+2)} = -1$
 $\left[\because \lim_{x \rightarrow 0^+} \frac{\tan^{-1} x}{x} = 1 \right]$
 $\lim_{x \rightarrow -2^+} f(x) = \lim_{x \rightarrow -2^+} \frac{|x+2|}{\tan^{-1}(x+2)}$
 $= \lim_{x \rightarrow -2^+} \frac{x+2}{\tan^{-1}(x+2)} = 1$
 $\therefore \lim_{x \rightarrow -2} f(x)$ does not exist.

2. $\lim_{x \rightarrow 0} \frac{(256+ax)^{1/8} - 2}{(32+bx)^{1/5} - 2} = f(0)$

Use L-Hospital rule

$$\Rightarrow \lim_{x \rightarrow 0} \frac{\frac{1}{8}(256+ax)^{-7/8} \cdot a}{\frac{1}{5}(32+bx)^{-4/5} \cdot b} = f(0)$$

On simplify, $\frac{a}{b} = \frac{64}{5} \cdot f(0)$

3. Use L-Hospital rule and Equating $Nr = 0$
 then, $3a - b - 2 = 0 \rightarrow (1)$
 $\Rightarrow 3a - 4b - 1 = 0 \rightarrow (2)$

Solving (1) and (2), $a = \frac{7}{9}, b = \frac{1}{3}$

Solving (1) and (2), $a = \frac{7}{9}, b = \frac{1}{3}$

4. $\lim_{x \rightarrow 0^+} f(x) = f(0)$
 $\Rightarrow \lim_{x \rightarrow 0^+} \left(1 + \frac{ax + bx^3}{x^2} \right)^{1/x} = 3$

$$\Rightarrow \lim_{x \rightarrow 0} \left(1 + \frac{a}{x} + bx \right)^{1/x} = 3$$

L.H.S. is exist when $a = 0$

$$\Rightarrow \lim_{x \rightarrow 0} (1 + bx)^{1/x} = 3$$

$$\Rightarrow e^6 = 3 \Rightarrow b = \log_e^3$$

5. For f to be continuous at $x = 0$

$$\begin{aligned} f(0) &= \lim_{x \rightarrow 0} f(x) \\ &= \lim_{x \rightarrow 0} \frac{\log(1+x+x^2) + \log(1-x+x^2)}{\sec x - \cos x} \\ &= \lim_{x \rightarrow 0} \frac{\log((1+x^2)^2 - x^2)}{1 - \cos^2 x} \cos x \\ &= \lim_{x \rightarrow 0} \frac{\log(1+(x^2+x^4))}{\sin^2 x} \cos x \\ &= \lim_{x \rightarrow 0} \frac{\log(1+(x^2+x^4))}{x^2+x^4} \times \frac{x^2}{\sin^2 x} \\ &\quad \times (1+x^2) \cos x = 1 \end{aligned}$$

6. $k = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(\cos x) - \cos x}{(\pi - 2x)^3}$

$$\Rightarrow k = \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin(\cos x) - \cos x}{(\cos x)^3} \times \frac{\sin^3\left(\frac{\pi}{2} - x\right)}{8\left(\frac{\pi}{2} - x\right)^3}$$

$$\Rightarrow k = -\frac{1}{6} \times \frac{1}{8} = -\frac{1}{48}$$

$$\left[\therefore \lim_{x \rightarrow 0} \frac{\sin x - x}{x^3} = -\frac{1}{6} \right]$$

7. $f(x)$ is discontinuous at $x = 1$ or $x = -1$

8. f is continuous when $5x = x^2 + 6$

$$\Rightarrow x^2 - 5x + 6 = 0$$

$$\Rightarrow x = 2, 3$$

$$9. \lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \left(\frac{A+3}{x^2} - \frac{3}{2!} + \frac{3}{4!}x^2 - \frac{3}{6!}x^4 + \dots \right)$$

For this limit to exist, we must have $A+3=0$ and in that case, we have

$$\lim_{x \rightarrow 0} f(x) = -\frac{3}{2} \text{ Now,}$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} B \tan \left(\frac{\pi}{[x+3]} \right) = B \tan \frac{\pi}{3} = B\sqrt{3}$$

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x) = f(0)$$

$$10. \Rightarrow \lim_{x \rightarrow \frac{\pi}{2}} \frac{1 - \sin^3 x}{3 \cos^2 x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{b(1 - \sin x)}{(\pi - 2x)^2} = a$$

By L-Hospital rule

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}} \frac{+3 \sin^2 x \cdot \cos x}{+6 \cos x \sin x} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{-b \cos x}{2(\pi - 2x)(-2)} = a$$

$$\Rightarrow \lim_{x \rightarrow \frac{\pi}{2}} \frac{\sin x}{2} = \lim_{x \rightarrow \frac{\pi}{2}} \frac{b \sin x}{8} = a$$

$$\Rightarrow \frac{1}{2} = \frac{b}{8} = a \quad \therefore a = \frac{1}{2}$$

$$11. \lim_{x \rightarrow 9^+} f(x) = \lim_{x \rightarrow 9^-} f(x) = f(9)$$

$$\Rightarrow \lim_{x \rightarrow 9^+} \frac{a|x^2 - 15x + 56|}{x-8} = \lim_{x \rightarrow 9^-} \frac{x-[x]}{x-8} = b$$

$$\Rightarrow \lim_{x \rightarrow 9^+} \frac{a(x-7)(x-8)}{x-8} = \lim_{x \rightarrow 9^-} \frac{x-8}{x-8} = b$$

$$\Rightarrow 2a = 1 = b$$

$$12. \lim_{x \rightarrow -1^-} f'(x) = \lim_{x \rightarrow -1^+} f'(x)$$

$$\Rightarrow \lim_{x \rightarrow -1^-} 2bx = \lim_{x \rightarrow -1^+} 2ax - b$$

$$\Rightarrow -2b = -2a - b$$

$$\Rightarrow -b = -2a \Rightarrow 2a = b$$

$\Rightarrow f(x)$ is also continuous

$$\Rightarrow \lim_{x \rightarrow -1^-} bx^2 - a = \lim_{x \rightarrow -1^+} ax^2 - bx - 2$$

$$\Rightarrow b - a = a + b - 2 \Rightarrow 2a = 2$$

$$\Rightarrow a = 1 \Rightarrow b = 2$$

$$\text{Equation is } x^2 - 3x + 2 = 0$$

$$13. f(x) = \frac{\log(1+x^2)}{(x^2-25)(x^2-1)}$$

$$= \frac{\log(1+x^2)}{(x-5)(x+5)(x-1)(x+1)}$$

since $\log(1+x^2)$ is continuous on $(-\infty, \infty)$, so

f is continuous on $R \sim \{-5, -1, 1, 5\}$ which clearly contains the interval $[6, 10]$.

14. $f(x) = [3x]$ is discontinuous when $3x =$ an integer

$$\Rightarrow x = \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \frac{4}{3}, \frac{5}{3}, \frac{6}{3}, \frac{7}{3}, \frac{8}{3} \in (0, 3)$$

$$15. f(x) = \begin{cases} 0 & ; 0 \leq x < 1 \\ x & ; 1 \leq x < 2 \\ 2x & ; 2 \leq x < 3 \\ 3(x-1) & ; 3 \leq x < 4 \\ 12 & ; x = 4 \end{cases}$$

16. $[x]$ is not continuous & differentiable at integral values (points) so $f(x)$ continuous

and differentiable in $(7, 9)$ if $\left[\frac{(x-5)^3}{A} \right] = 0$

$$\Rightarrow A \geq (9-5)^3$$

$$\Rightarrow A \geq 64 \quad \therefore A \in [64, \infty)$$

CONTINUITY

17. $f(x)$ is continuous when $x^2 - ax + 3 = 2 - x$
 $\Rightarrow x^2 - (a-1)x + 1 = 0$. This must have two distinct roots $\Rightarrow \Delta > 0$
 $\Rightarrow (a-1)^2 - 4 > 0$
18. $x \in [1, 4)$. The value of $f(x)$ is zero
 $x \in [4, 2\pi)$. The value of $f(x)$ is -1.
19. It is discontinuous at $x = 0, 2, 3$
20. It is everywhere continuous.
21. $x^2 + kx + 1 \geq 0$ and $x^2 - k$ must not have any real root
 $\therefore k^2 - 4 \leq 0$ & $k < 0$
 $\Rightarrow k \in [-2, 2]$ and $k < 0$
 $\Rightarrow k \in [-2, 0)$
22. $f(x)$ is discontinuous when $x^3 = n; n \in \mathbb{Z}$
 $\Rightarrow x = n^{\frac{1}{3}}$
 $f\left(\frac{3}{2}\right) = (-1)^3 = -1$

$$\left. \begin{array}{l} \text{for } x \in (-1, 0) f(x) = (-1)^{-1} = -1 \\ \text{for } x \in [0, 1) f(x) = (-1)^0 = 1 \end{array} \right\} \Rightarrow f^l(x) = 0$$
23. $|x|$ is not differentiable at $x = 0$
 $|x|$ is continuous at $x = 0$
24. $f(x) = \cos^{-1}(\cos x) = \begin{cases} 2\pi + x & \text{for } -2\pi < x < -\pi \\ -x & \text{for } -\pi \leq x < 0 \\ x & \text{for } 0 \leq x \leq \pi \\ 2\pi - x & \text{for } \pi < x < 2\pi \end{cases}$
 $f(x)$ is continuous at $x = \pi, -\pi$
25. Clearly, $f(x) = 0$ for each integral value of x .
 Also, if $0 < x < 1$, then $0 < x^2 < 1$,
 $\Rightarrow [x] = 0$ and $[x^2] = 0$
 $\therefore f(x) = 0$ for $0 < x < 1$
 Again, if $1 \leq x < \sqrt{2}$ then $1 \leq x^2 < 2$
 $\Rightarrow [x] = 1, [x^2] = 1$
 $\therefore f(x) = 0$ for $1 \leq x < \sqrt{2}$
 $\therefore f(x) = 0$ is continuous at $x = 1$

JEE-MAIN-JR-MATHS VOL-IV

However, at points x other than integers and not lying between 0 and $\sqrt{2}$, $f(x) \neq 0$

26. $u = \frac{2}{x-2}$ is discontinuous at $x = 2$
 $f(u) = \frac{1}{u^2 - 17u + 66} = \frac{1}{(u-11)(u-6)}$ is discontinuous at $u = 6, 11$
 $\therefore \frac{2}{x-2} = 6, 11 \Rightarrow x = \frac{7}{3}, \frac{24}{11}$
27. $f(x) = |2 \operatorname{sgn}(2x)| + 2$
 $= \left| 2 \cdot \frac{2x}{|2x|} \right| + 2 = \left| 2 \cdot \frac{x}{|x|} \right| + 2$
 $f(0) = 2$
 $f(0^+) = |2 \times 1| + 2 = 4$
 $f(0^-) = |2(-1)| + 2 = 4$
 $f(0^+) = f(0^-) \neq f(0)$
 $\therefore f(x)$ has removable discontinuity
28. $f(x)$ is discontinuous when $x^3 - x = 0$
 $\Rightarrow x = 0, -1, 1$
29. Function continuous for all x
 $\Rightarrow 2 \sin x + a \neq 0 \Rightarrow \sin x \neq \frac{-a}{2}$
 $\Rightarrow \left| \frac{a}{2} \right| > 1 \Rightarrow a < -2 \text{ (or) } a > 2$
30. $[x] \sin \pi x$ is continuous for every real x .
31. Use L-Hospital's rule.
32. If
 $f(x) = \lim_{x \rightarrow \infty} (\sin^2 x)^n$
 $= \begin{cases} 1, & \text{if } x = (2n+1)\frac{\pi}{2}, \forall n \in \mathbb{Z} \\ 0, & \text{if } x \neq (2n+1)\frac{\pi}{2}, \forall n \in \mathbb{Z} \end{cases}$



LEVEL-IV

1. Assertion(A): $f(x) = \frac{\sin \{[x]\pi\}}{1+x^2}$ is continuous on $\mathbb{R}$. where $[x]$ & $\{x\}$ denotes greatest integer function and fractional part of x .

Reason (R): Every constant function is continuous on $\mathbb{R}$

- 1) Both A and R are true and R is the correct explanation of A
- 2) Both A and R are true and R is not the correct explanation of A
- 3) A is true but R is false
- 4) R is true but A is false

2. Assertion(A): $f(x) = x \left(\frac{1+e^{1/x}}{1-e^{1/x}} \right)$, ($x \neq 0$),

$f(0) = 0$ is continuous at $x = 0$.

Reason(R) : A function is said to be continuous at 'a' if both limits are exists and equal to $f(a)$.

- 1) Both A and R are true and R is the correct explanation of A
- 2) Both A and R are true and R is not the correct explanation of A
- 3) A is true but R is false
- 4) R is true but A is false

3. Match the following :

List - I List - II
p) The no. of points of discontinuity of the function 1) 0

$$f(x) = \frac{1}{\log |x|} \text{ is}$$

q) If $f(x) = (x+1)^{\operatorname{cosec} x}$ 2) 3
is continuous at $x=0$
then $f(0)=$

r) $f(x) = \frac{\cos x - \sin x}{\cos 2x}$, 3) 1

$x \neq \frac{\pi}{4}$ If f is continuous on 4) e

$$\left[0, \frac{\pi}{4}\right] \text{ then } f\left(\frac{\pi}{4}\right) =$$

s) If $f(x) = \frac{\sqrt{1+\sin x} - \sqrt{1-\sin x}}{x}$ 5) $\frac{1}{\sqrt{2}}$ is

continuous everywhere then $f(0) =$

Then the correct match for List - I from List - II

p	q	r	s
1) 1	2	3	4
2) 4	3	1	2
3) 2	4	5	3
4) 2	4	1	5

4. List - I List - II

p) $f(x) = [x]$ is continuous on 1) $\mathbb{R}$

q) $f(x) = |x|$ is continuous on 2) $\mathbb{R} - \mathbb{Z}$

r) $f(x) = \frac{|x-2|}{x-2}$ is continuous on 3) $\mathbb{R} - \{2\}$

Then the correct match for List - I from List - II

p	q	r
1) 1	2	3
2) 1	3	2
3) 2	1	3
4) 2	3	1

5. Assertion(A): $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$ is continuous at $x = 0$

Reason(R):

$$\text{Both } h(x) = x^2, g(x) = \begin{cases} \sin\left(\frac{1}{x}\right), & \text{if } x \neq 0 \\ 0, & \text{if } x = 0 \end{cases}$$

are continuous at $x = 0$

- 1) Both A and R are true and R is the correct explanation of A
- 2) Both A and R are true and R is not the correct explanation of A
- 3) A is true but R is false 4) R is true but A is false

LEVEL-IV-KEY

1) 1 2) 1 3) 3 4) 3 5) 3

LEVEL-IV-HINTS

1. A is true, R is true and $R \Rightarrow A$
 $\sin n\pi = 0$, for $n \in \mathbb{Z}$
2. $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$
3. Simplify
4. $[x]$ is continuous on $\mathbb{R} - \mathbb{Z}$
 $|x|$ is continuous on $\mathbb{R}$
5. A is true but R is not correct
because, $\lim_{x \rightarrow 0} g(x)$ does not exist